

Mitochondrial viability – Triphenyltetrazolium chloride (TTC) assay

Materials:

1. Spinal cord fragment
2. Tissue cutter
3. Tap water
4. Distilled water
5. Incubator chamber
6. Refrigerator
7. Micro tubes
8. 2 % TTC solution

Procedures:

1. After euthanasia, collect the spinal cord fragment from the lesion epicenter and subject to an a refrigerated artificial cerebrospinal fluid (ACSF) for 5 min and slice (400- μ M thick) with the aid of a tissue cutter.
2. After processing in the perfusion chamber, incubate the slices in a 1.5 ml of 2 % TTC solution at 37 °C for 90 min.
3. Remove TTC after this period, and wash the tissue with 0.9 % saline solution.
4. Add 1.5 mL of a 1:1 dimethylsulfoxide (DMSO)/ethanol solution, and incubate at room temperature in the dark for 48 h, to solubilize formazan.
5. Read the absorbance in a spectrophotometer at 485 nm, and the results should be normalized by tissue weight.